Chapter -1

Computer Networking

Network:

A network is a system that connects two or more devices, such as computers, mobile phones and smart machine, allowing them to share information, communicate and work together.

Internet is a type of network and www is the server of internet.

Element	Feature	Example
Message (data)	The data exchanged between sender and receiver.	Text, audio, video, picture
Sender	The device that sends the message.	Computer, mobile phone, network node
Receiver	The device that receives the message.	Computer, mobile phone
Communication Medium	The physical or wireless path for the message to travel.	Twisted pair cables, fibre optics, satellite
Protocol	A set of rules that control the communication between sender and receiver.	TCP/IP, HTTP, Bluetooth protocol

IMPORTANCE OF NETWORKS

- **Telemedicine:** Doctors consult with patients in remote villages via secure video platforms.
- **Online learning:** Virtual classrooms like Microsoft Teams or Zoom allow real-time learning.
- **Cloud storage:** Files are saved in data centres and are accessible from any device (e.g., Google Drive).
- **Smart cities:** Traffic signals, cameras and emergency systems are linked through IoT networks.
- **Autonomous systems:** Self-driving cars exchange data through low-latency 5G networks.

Networking Media:

Networking media means the path through which data travels in a network. It is also known as transmission media. Networking Media is of two types.

- Wired Media
- Wireless Media.

Wired Media: In this media cables are used to connect devices in a network. Example of wired media are:

- Twisted pair cable
- Coaxial cable
- Optical fibre cable

Wireless Media:

In this media data is sent through the air using signals. It is also known as unguided media.

Examples of wireless media:

Bluetooth

Infrared Waves

Wi-Fi

Types of Networking:

There are four type of Networking

PAN (Personal Area Network):-

In this network devices are very close to each other, usually within a few meters. Used to connect personal devices.

e.g. Connecting laptop with mobile

Connecting smartphone to smartwatch using Bluetooth.

LAN (Local Area Network):

Connects computers and devices within a small area, like a home, a classroom or office. It allows user to share files, printers and internet.

e.g. : School computer lab

Laptop, TVs and phones connected to a home Wi-Fi router.

MAN (Metropolitan Area Network):

Connects computers and devices within city or a large town.

e.g. CCTV cameras and smart traffic lights across the city using fibre optic cable.

WAN (Wide Area Network)

Connects computers and devices within a large distance from one city or country to another city or country. Used for global communication.

e.g. Internet.

Network Topology:

Network Topology is the way in which computers and other devices are connected.

Star Topology
All devices are connected to a central device like a hub or switch.

Advantages and disadvantages of **Star Topology**:

Advantages of Star Topology	Disadvantages of Star Topology
Easy to install and manage	Requires more cable compared to other topologies
Failure of one device does not affect the whole network	If the central hub/switch fails, entire network stops
Easy to detect and fix faults	Costly due to hub/switch and wiring
Easy to add or remove devices	Requires more maintenance of central device

Ring Topology
Devices are connected in a circular form, where each device is connected to two others.

Bus Topology
All devices are connected to a single central cable (bus).

Advantages and disadvantages of **Bus Topology**:

Advantages of Bus Topology	Disadvantages of Bus Topology
Simple and easy to install	If the main cable fails, entire network stops
Requires less cable than star topology	Difficult to identify faults
Cost-effective (low setup cost)	Performance decreases with more devices
No need for central device	Limited cable length and number of devices

Mesh Topology
Every device is connected to every other device in the network.

Advantages and disadvantages of **Mesh Topology**:

Advantages of Mesh Topology	Disadvantages of Mesh Topology
Highly reliable (multiple paths for data)	Very expensive (requires more cables and devices)
Failure of one link does not affect the network	Complex to install and manage
High security and privacy	Requires a large amount of wiring

Networking Devices

Modem
Converts digital signals into analog signals and vice versa to connect to the internet. This process is known as **modulation and demodulation**.

Hub
A basic device that connects multiple computers and sends data to all devices.

Switch
Connects devices in a network and sends data only to the intended device (more efficient than hub).

Router
Connects different networks and directs data between them (used for internet connection).

Gateway
Connects different types of networks and acts as a translator.

Repeater
Boosts or regenerates signals to extend network distance.

cloud router is a virtual router that runs in a cloud environment instead of physical hardware.

DNS (Domain Name System)

DNS is a system that converts human-readable domain names into IP addresses so that computers can communicate with each other.

Example:
When you type **www.google.com**, DNS converts it into an IP address like **142.250.xxx.xxx**.

Network Protocol

A **network protocol** is a set of rules that define how data is transmitted and communicated between devices in a network.

Examples:
IP (Internet Protocol)
Responsible for assigning addresses and delivering packets from source to destination.

TCP (Transmission Control Protocol)
Ensures reliable and error-free delivery of data.

HTTP (HyperText Transfer Protocol)
Used to transfer web pages on the internet.

Cyber Security and Network Safety

Cyber security and network safety refer to the protection of computers, networks, and data from unauthorized access, attacks, and damage.

End-to-End Encryption (E2EE):
Ensures that data is encrypted at the sender's end and can only be decrypted by the receiver, preventing unauthorized access.

VPN (Virtual Private Network):
Creates a secure, encrypted connection over the internet to protect data and maintain privacy.

MFA (Multi-Factor Authentication):
Adds an extra layer of security by requiring two or more verification methods (e.g., password + OTP).

AI-Based Threat Detection:
Uses artificial intelligence to identify and respond to cyber threats in real time by analyzing patterns and unusual activities.